CLASS 9 — SCIENCE · EASY LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Sodium ion is: 1

(a) Na^{+} **(b)** Na^{2-} **(c)** Na^{2+} **(d)** Na^{-}

2. The electronic configuration of carbon is: 1

(a) 2, 4 **(b)** 4, 2 **(c)** 2, 6 **(d)** 2, 2

3. 120 seconds equals: 1

(a) 2 minutes **(b)** 1 minute **(c)** 12 minutes **(d)** 20 minutes

4. The percentage of solute in 10 g salt + 90 g water: 1

(a) 90 per cent **(b)** 100 per cent **(c)** 10 per cent **(d)** 9 per cent

5. Ionic compounds are usually: 1

(a) gases **(b)** low melting liquids
(c) solids with high melting point **(d)** soft solids

6. Hydrogen has: 1

(a) three isotopes **(b)** no isotope **(c)** four isotopes **(d)** two isotopes

7. SI unit of time: 1

(a) minute **(b)** day **(c)** hour **(d)** second

8. Sand in water forms a: 1

(a) solution **(b)** suspension **(c)** colloid **(d)** compound

9. Atoms are the building blocks of: 1

(a) only metals **(b)** energy **(c)** matter **(d)** only gases

10. The nucleus contains: 1

(a) electrons only **(b)** only protons
(c) protons and neutrons **(d)** only neutrons

11. The symbol for the speed of light: 1

(a) c **(b)** v only **(c)** l **(d)** s

12. Separation of iron filings uses a: 1

(a) magnet **(b)** sieve **(c)** filter **(d)** funnel

13. Chloride ion is: 1

(a) Cl^{2+} **(b)** Cl^{2-} **(c)** Cl^{-} **(d)** Cl^{+}

14. The electronic configuration of oxygen is: 1

(a) 2, 8 **(b)** 2, 6 **(c)** 2, 5 **(d)** 2, 4

15. A thermometer measures: 1

(a) temperature **(b)** time **(c)** mass **(d)** length

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Name the bond formed by transfer of electrons. 2

17. Write the electronic configuration of sodium. 2

18. State the SI unit of length and mass. 2

19. What is a solution? 2

20. State the law of constant proportions. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the mass of CO2 formed when 12 g of carbon burns in 32 g of oxygen and name the law used. 3

22. Write three conclusions from Rutherford's experiment. 3

23. Estimate the number of minutes in a day. 3

24. Find the mass of salt in 200 g of 5 per cent solution. 3

OR

Write the atomicity of: He, O_3, P_4, S_8, H_2, CH_4. 3

25. Write the electronic configurations of F, Ne, Na. 3

OR

What are the sources of error in measurement? Give three. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the use of models in science with three examples. 5

OR

Describe how to separate a mixture of sand, salt and iron filings. 5

27. State and explain the law of constant proportions with an example. 5

OR

Describe Rutherford's model of the atom. 5

END OF QUESTION PAPER — **Total: 50 Marks**